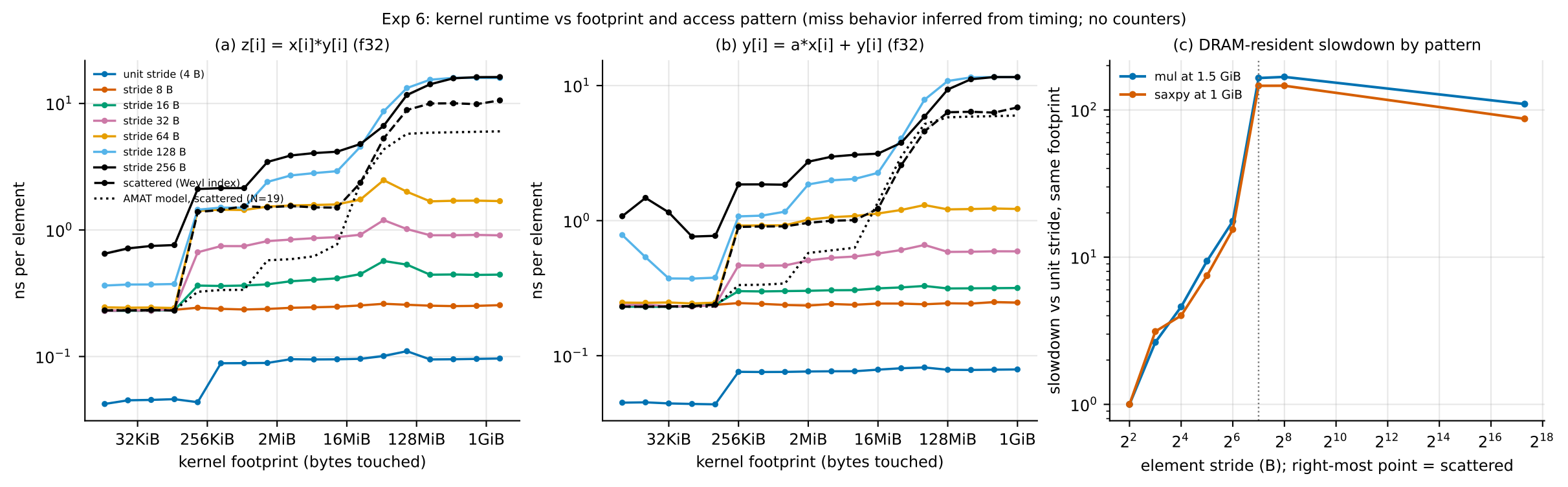
Exp 6: kernel runtime vs footprint and access pattern (miss behavior inferred from timing; no counters)
(a) z[i] = x[i]*y[i] (f32)
(b) y[i] = a*x[i] + y[i] (f32)
(c) DRAM-resident slowdown by pattern
unit stride (4 B)
stride 8 B
stride 16 B
stride 32 B
stride 64 B
stride 128 B
stride 256 B
scattered (Weyl index)
AMAT model, scattered (N=19)
ns per element
kernel footprint (bytes touched)
32KiB
256KiB
2MiB
16MiB
128MiB
1GiB
mul at 1.5 GiB
saxpy at 1 GiB
slowdown vs unit stride, same footprint
element stride (B); right-most point = scattered